NAKSHATRA
THE AEROSPACE HACKATHON 2026

ORBITAL FUTURE DECISION ENGINE

FANTASTIC 4

UNCERTAINTY-AWARE SPACE DEBRIS TRACKING, FUTURE SCENARIO ANALYSIS & DECISION SUPPORT

Aerospace Hackathon 2026 · Software-Only Prototype



STATUS: ANALYSIS · DATA: TLE | MISSION / ORBITAL DECISION SUPPORT

OBJECT: SAT-X · MODE: SIMULATION

Software

No hardware. A software-only decision-support prototype.

THEME **Space Debris Detection, Tracking & Mitigation**

React + TypeScript
Frontend

CesiumJS
3D Orbital View

Python + FastAPI
Backend

NumPy / SciPy
Numerics

TLE Propagation
Astrodynamics

PostgreSQL
Data Store

PROTOTYPE SCOPE

Physics-grounded, uncertainty-aware decision support.

NOT autonomous.
Human operator stays in the loop.

PROBLEM STATEMENT

OBJECT: SAT-X · THREAT: DEBRIS-07

The Orbital Safety Challenge

SAT-X
Operational spacecraft

DEBRIS-07
Crossing object

CLOSE APPROACH
Minimum distance event

ALERT
Single risk value

WHAT HAPPENS AFTER THE ALERT?

A risk number alone doesn't show:

- How uncertain the prediction is
- What future states are plausible
- What happens after a maneuver
- Whether another object becomes a threat
- How stable the decision is

PROPOSED SOLUTION

Our Solution

01
TRACK
Orbital data

›

02
PREDICT
Propagation + Risk

›

03
EXPLORE
Future-State Envelope

›

04
COMPARE
No Action / A / B

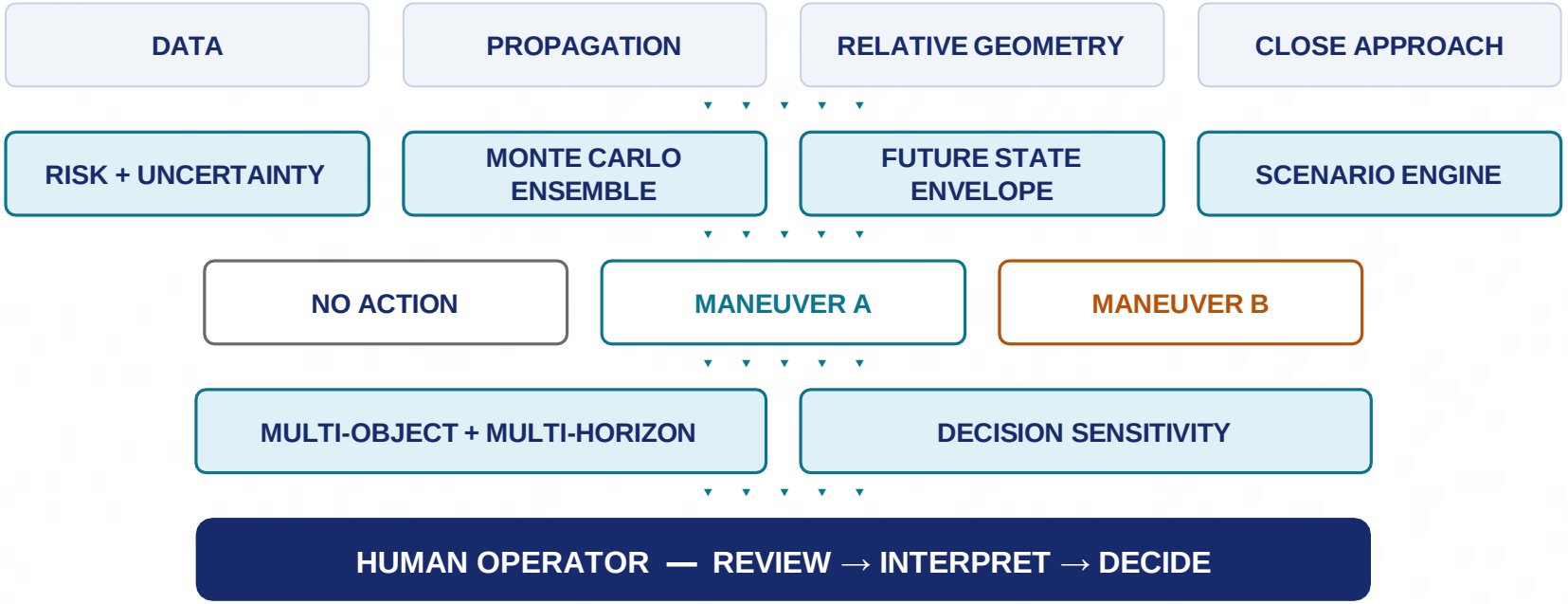
›

05
UNDERSTAND
Multi-Object + Sensitivity

A decision-support layer built on established orbital science.

SYSTEM ARCHITECTURE

Processing Pipeline



HOW IT WORKS

From Alert to Future Decision

TYPICAL SIMPLIFIED WORKFLOW

- Track
- Propagate
- Close Approach
- Risk
- Alert

OUR PROTOTYPE

DECISION LAYER

- **Track**
- **Risk + Uncertainty**
- **Future State Envelope**
- **Scenario Comparison**
- **Multi-Object Consequences**
- **Multi-Horizon Exposure**
- **Threat Transfer**
- **Decision Sensitivity**

CAPABILITIES

Features & Novelty

01

ORBITAL TRACKING

Established propagation models

02

CLOSE-APPROACH ANALYSIS

Distance · TCA · Rel. velocity

03

FUTURE STATE ENVELOPE

Ensemble under uncertainty

04

SCENARIO EXPLORER

No Action · A · B

05

MULTI-HORIZON

24H · 7D · 30D

06

DECISION SENSITIVITY

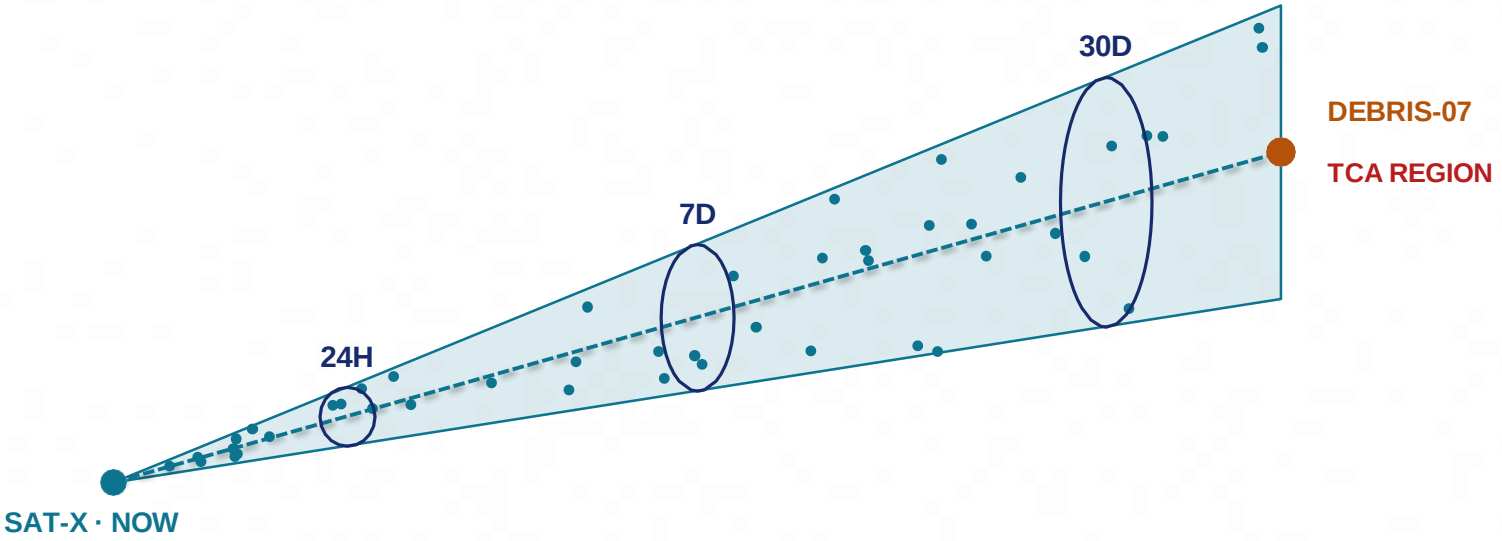
Stability under uncertainty

Extends established debris tracking into an uncertainty-aware, human-in-the-loop decision framework.

SIGNATURE CONCEPT

OBJECT: SAT-X · HORIZON: 30 DAYS

Future State Envelope

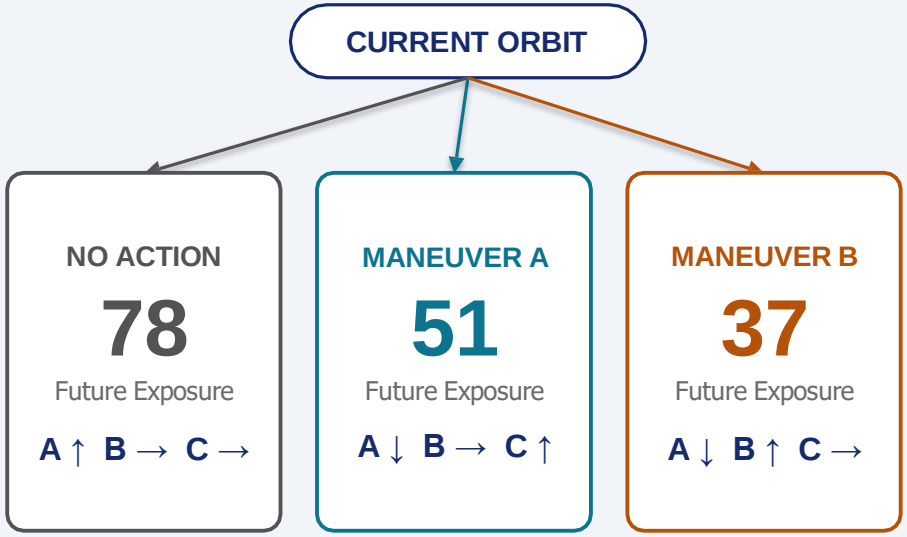


“One predicted path → a distribution of plausible futures.”

SCENARIO COMPARISON

ILLUSTRATIVE DEMO DATA

Branching Futures



Each scenario is evaluated against Debris A · B · C independently.

THREAT TRANSFER

BEFORE

Debris A → HIGH

Debris B → LOW

▼

AFTER MANEUVER

Debris A → LOWER

Debris B → HIGHER

A maneuver that reduces one modeled threat can increase another.

DECISION METRICS

Exposure & Sensitivity

ILLUSTRATIVE PROTOTYPE METRIC

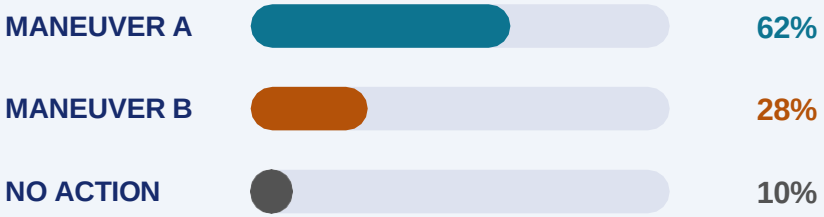
FUTURE EXPOSURE INDEX (FEI)

78	51	37
NO ACTION	MANEUVER A	MANEUVER B

24H + 7D + 30D · Multiple Objects

Illustrative prototype metric — not an industry standard.

DECISION SENSITIVITY



Decision Sensitivity: 38%

How stable is the decision when uncertainty changes?

Illustrative simulation ensemble — not real spacecraft probabilities.

ENGINEERING HONESTY

Limitations & Next Steps

DATA QUALITY

Freshness + uncertainty affect results

MODEL ASSUMPTIONS

Prototype metrics require validation

SIMPLIFIED MANEUVERS

Conceptual Δv — not flight-qualified

COMPUTATIONAL SCALE

Needs optimization at catalog scale

Decision support — not autonomous spacecraft control.

THE TEAM

Fantastic 4

AS	SG	NC	CL
Anuj Singh	**Sarthak Gavnekar**	**Nitin Chherke**	**Chirag Lalwani**
Team Leader	Team Member	Team Member	Team Member

FROM ONE PREDICTED FUTURE

→ TO PLAUSIBLE FUTURES

THANK YOU

Explore plausible futures. Understand the consequences. Support better decisions.

FANTASTIC 4 · ORBITAL FUTURE DECISION ENGINE

Technical R&D Dossier — QR Code

For detail research of team - https://orbital-future-anuj-singh.netlify.app